

HypeLoRA: Hyper-Network-Generated LoRA Adapters for Calibrated Language Model Fine-Tuning

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Abstract. Modern Transformer-based models frequently suffer from miscalibration, producing overconfident predictions that do not reflect true empirical frequencies. This work investigates the calibration dynamics of LoRA: Low-Rank Adaptation and a novel hyper-network-based adaptation framework as parameter-efficient alternatives to full fine-tuning for RoBERTa. Evaluating across the GLUE benchmark, we demonstrate that LoRA-based adaptation consistently achieves calibration parity with (and in specific tasks exceeds) full fine-tuning, while maintaining significantly higher parameter efficiency. We further explore a dynamic approach where a shared hyper-network generates LoRA factors (A and B matrices) to induce structural coupling across layers. This approach produced results similar to standard LoRA fine-tuning, even achieving better MCC on CoLA dataset. Our study also reveal a critical trade-off: constraining the adaptation space (e.g., freezing matrices A) acts as a powerful regularizer that enhances Expected Calibration Error (ECE), but necessitates a carefully balanced sacrifice in downstream task accuracy. To support future research, we provide a unified and reproducible implementation of contemporary calibration metrics, including ECE, MCE, and ACE. Our findings clarify the relationship between parameter efficiency and probabilistic reliability, positioning structured low-rank updates as a viable foundation for uncertainty-aware Transformer architectures. <https://github.com/btrojan-official/HypeLoRA>

Keywords: Fine-tuning · LoRA · Hyper-network · RoBERTa · Calibration · ECE

1 Introduction

Transformer-based architectures have become the dominant paradigm across a wide range of machine learning domains, including natural language processing [4], computer vision [5], and speech recognition [1]. While these models achieve state-of-the-art predictive accuracy, it is now well established that their probabilistic outputs are often poorly calibrated [3]. Formally, a model is considered calibrated if the predicted confidence of a given class matches the true

empirical frequency of that class among all samples assigned that confidence score — that is, among all inputs where the model outputs probability p , exactly a fraction p should truly belong to the predicted class.

In this work, we explore parameter-efficient alternatives for calibrating transformer based encoders. We evaluate fine-tuned and augmented on LoRA RoBERTa model on the GLUE benchmark using a comprehensive set of calibration metrics, including Expected Calibration Error (ECE), Adaptive Calibration Error (ACE), and Maximum Calibration Error (MCE). In addition, we investigate the use of hyper-networks as a mechanism for inducing dynamic calibration behavior within frozen transformer architectures.

Additionally, we study a framework in which a hyper-network generates low-rank calibration signals. This setup aims to induce structural coupling across layers, forcing the model to learn a global calibration strategy rather than independent adjustments. We anticipated that this shared generation process would act as a regularizer, tempering overly sharp probability distributions. While this approach ultimately fails to yield consistent improvements, it provides valuable insight into the limitations of hyper-network-driven calibration for transformers.

Beyond empirical evaluation, we also consolidate and implement a unified set of recent calibration metrics, providing clear and reproducible reference implementations. This addresses the current fragmentation in calibration evaluation practices and facilitates more systematic comparison across methods.

Our contributions can be summarized as follows:

- We provide an evaluation of standard and LoRA fine-tuning for calibration of RoBERTa models on the GLUE benchmark using multiple complementary calibration metrics i.e. ECE, MCE, ACE.
- We investigate the use of hyper-networks and LoRA as mechanisms for systematic model calibration and for analyzing the underlying causes of model failure.
- We implement modern calibration metrics in a single evaluation framework.

2 Related Work

Pre-training. Large-scale self-supervised pre-training on generic corpora yields representations that transfer across downstream tasks, as demonstrated by masked language modeling and autoregressive objectives [4,16]. Domain- and task-adaptive pre-training on unlabeled data can further improve performance [7], yet the high computational cost of pre-training large backbones motivates parameter-efficient adaptation methods.

Fine-tuning large language models. Full-parameter fine-tuning is the most direct adaptation approach, but is computationally and memory demanding for large models [17]. When a single backbone such as RoBERTa [16] must be tailored to many tasks, storing full task-specific parameters and optimizer states becomes infeasible, motivating more resource-efficient alternatives.

Parameter-efficient fine-tuning. PEFT methods update only a small parameter subset per task while achieving performance comparable to full fine-tuning.

Adapters [9] insert trainable bottleneck modules into each transformer block. Prefix-tuning [13] prepends learnable tokens to the input, optimizing a continuous prompt without modifying model weights. LoRA [10] injects trainable low-rank matrices into attention layers without any activation function, substantially reducing trainable parameter count and memory cost.

Model Calibration [25]. A calibrated model’s confidence estimates align with empirical likelihoods: formally, perfect calibration requires $P(Y = y|Q = q) = q$ over the joint distribution $P(Q, Y)$. We evaluate calibration primarily via expected calibration error (ECE) [6], alongside Classwise ECE, MCE, ACE, Thresholded ACE, and Brier Score [19,12,20,2]. Post-hoc methods such as temperature scaling [18], Platt scaling [22], and isotonic regression [26] are simple but globally applied and brittle under distribution shift. Training-time approaches like label smoothing [15] and confidence-aware regularization [14] are more expressive but require costly retraining.

Hyper-networks. Hyper-networks parameterize weights as functions of external inputs, enabling dynamic generation of adaptation parameters [8]. In PEFT, conditioning hyper-networks on task identity to generate adapter or low-rank update parameters improves parameter sharing and multi-task performance without per-task parameter duplication [21]. Generating full weight tensors remains challenging due to scalability and stability concerns [11], so practical approaches constrain generation to compact structures such as bottleneck adapters or low-rank factors.

3 Proposed Approach

We propose a parameter-efficient calibration mechanism that injects low-rank updates into a frozen RoBERTa encoder [16] via a compact hyper-network (Figure 1). The hyper-network conditions on a learned embedding associated with each target weight matrix, producing coordinated adaptation signals across all transformer layers while keeping all backbone parameters frozen.

3.1 Problem Definition

Let $f_\theta(x)$ denote a pretrained transformer encoder with frozen parameters θ , producing a probability distribution over C classes for input x :

$$p_\theta(y | x) = \text{softmax}(f_\theta(x)). \quad (1)$$

Our goal is to improve predictive calibration without modifying pretrained weights. Instead of post-hoc logit adjustments (e.g., temperature scaling), we introduce structured low-rank perturbations inside the transformer layers.

3.2 Layer-Conditioned Low-Rank Updates with Hyper-Network

For a weight matrix $W \in \mathbb{R}^{d \times d}$ of pretrained transformer, we apply a LoRA-style [10] low-rank perturbation:

$$W' = W + \alpha AB, \quad (2)$$

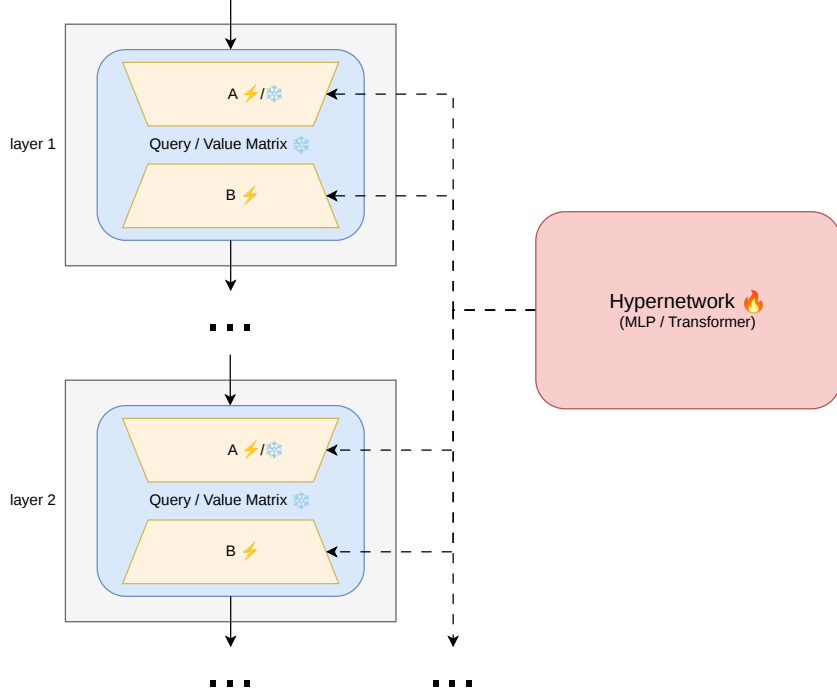


Fig. 1. A hyper-network generates the weights for the Query and Value matrices in all attention blocks of the RoBERTa model, while the original pretrained weights remain frozen. The figure illustrates the approach in which the hyper-network produces both the A and B matrices. In this work, we also present a variant where only the B matrices are generated by the hyper-network, and A matrices are fixed with randomly initialized values, which isn't shown on this figure.

where $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$ are low-rank factors of rank $r < d$, and α is a fixed scaling coefficient. We apply this update to the Query and Value projection matrices in each attention block.

Unlike standard LoRA, which trains all of A and B matrices independently, we generate these factors via a shared hyper-network H_ϕ , conditioned on layer embedding. Each target weight matrix — specifically the Query and Value projections in each layer — is associated with a dedicated learned embedding $e \in \mathbb{R}^{d_h}$. Concretely, if both A s and B s are generated for the Query and Value matrices in layer ℓ , there are four embeddings per layer: $e_A^Q[\ell]$, $e_B^Q[\ell]$, $e_A^V[\ell]$, $e_B^V[\ell]$. Hyper-network H_ϕ maps each such embedding e to target low-rank factor:

$$A_\ell^Q = H_\phi(e_A^Q[\ell]) \quad B_\ell^Q = H_\phi(e_B^Q[\ell]) \quad A_\ell^V = H_\phi(e_A^V[\ell]) \quad B_\ell^V = H_\phi(e_B^V[\ell]) \quad (3)$$

The resulting vectors are reshaped into matrices of appropriate dimensions and applied to their respective weight matrices. This ties all layer adaptations through a single generator, enforcing structural coherence and drastically reducing parameter count relative to per-layer LoRA. The architecture of H_ϕ is either a lightweight MLP or a small Transformer encoder operating over all embeddings jointly; implementation details are given in Section 4.3.

Variants. We consider two operating modes depending on whether A matrices are generated or fixed:

- **Full generation:** all A and B matrices are produced by H_ϕ , meaning that there are four embeddings per layer in this scenario.
- **Fixed-A:** A matrices are initialized once from a Kaiming uniform distribution (similarly to [10]) and held fixed throughout training; only B matrices are generated by H_ϕ , meaning that there are only two embeddings per layer. The intuition is that fixed random A matrices act as structured noise injection into the frozen base model, encouraging the learned B matrices to be more robust.

3.3 Training and Inference

During training, H_ϕ generates A s and B s for all target weight matrices. They are applied transiently in the forward pass — the stored pretrained weights are never modified. Gradients flow through the low-rank projections back to H_ϕ , whose parameters ϕ are updated while θ -original pretrained model weights-remains frozen. We optimize the standard cross-entropy loss.

At inference, the hyper-network again produces the low-rank factors on the fly, or they can be precomputed once to reduce runtime overhead. This design maintains a strict separation between the frozen backbone and the learned adaptation, ensuring a minimal memory footprint and no risk of corrupting the pretrained representations.

4 Experimental Setup

4.1 Datasets

We evaluate all of the experiments on the General Language Understanding Evaluation (GLUE) benchmark, a collection of sentence- and sentence-pair language understanding tasks designed to provide a standardized comparison of model performance across diverse NLP capabilities [24]. In this work, we use only GLUE tasks formulated as classification problems.

- **CoLA (Corpus of Linguistic Acceptability).** A single-sentence acceptability task where the model predicts whether a sentence is linguistically acceptable (binary). CoLA has 8.5k training examples and 1k test samples.
- **SST-2 (Stanford Sentiment Treebank).** A single-sentence sentiment classification task where the model predicts positive vs. negative sentiment (binary). SST-2 has 67k training examples and 1.8k test examples.

- **QNLI (Question-answering Natural Language Inference)**. A sentence-pair task (QA/NLI) where the model predicts whether a context sentence contains the answer to a question (binary). QNLI has 105k training examples and 5.4k test examples.
- **MRPC (Microsoft Research Paraphrase Corpus)**. A sentence-pair paraphrase task where the model predicts whether two sentences are paraphrases (binary). MRPC has 3.7k training examples and 1.7k test examples.
- **RTE (Recognizing Textual Entailment)**. A sentence-pair inference task (NLI) where the model predicts whether a hypothesis is entailed by a premise (binary). RTE has 2.5k training and 3k test examples.
- **MNLI (Multi-Genre Natural Language Inference)**. A sentence-pair natural language inference task where the model predicts the relationship between a premise and a hypothesis (entailment, contradiction, or neutral; three-way classification). MNLI has 393k training examples and 20k test examples.

4.2 Training and Evaluation Setup

Unless stated otherwise, we follow the experimental protocol from the original LoRA work [10] for training and evaluation. The two main differences are (i) the introduction of a hyper-network to generate the low-rank update parameters and (ii) the specific injection strategy used to apply these generated updates within the frozen RoBERTa encoder [16]. For our approach, we use different peak learning rates (e.g., $1e-5$ for the MLP and $4e-4$ for the Transformer), as we observed that lower learning rates lead to more stable and effective training when using the MLP-based hyper-network.

4.3 Hyper-network Architectures

We consider two hyper-network architectures for generating the LoRA factors: a multilayer perceptron (MLP) and a Transformer encoder [23]. In both cases, each transformer layer identifier ℓ is represented by a learnable embedding of dimension 128.

MLP. The MLP consists of four fully connected layers. The input dimension is 128, all hidden layers have width 2048, and the GELU as its activation function. The output layer projects to the flattened LoRA parameter space. In all experiments, the hidden size of RoBERTa is 768 and the LoRA rank is $r = 8$, so the output dimension corresponds to 768×8 per generated factor.

All MLP weights are initialized from a normal distribution. In the configuration where A matrices are fixed, it is initialized once using Kaiming uniform initialization and kept frozen, while only B is generated by the hyper-network.

Transformer. In the Transformer-based variant, all layer embeddings (dimension 128) are first projected with a linear layer to a hidden dimension of 256 and then processed jointly by a Transformer encoder with 2 layers, 16 attention heads, and model dimension 256. The outputs corresponding to each

layer are passed through a final linear projection to produce the flattened LoRA parameters, either for both A and B matrices or only for B s when A s are fixed.

All learnable parameters are initialized from a normal distribution, except for the fixed A matrices, which use Kaiming uniform initialization.

4.4 Metrics

Exact calibration measurement with finite samples is not possible due to continuous predicted confidence values. In practice, calibration error is approximated by partitioning N predictions into M bins $\{b_1, \dots, b_M\}$ based on predicted probabilities and comparing average confidence with empirical accuracy within each bin. $M = 10$ for all of our experiments.

The most widely used metric is **Expected Calibration Error (ECE)** [19], defined as the weighted average of per-bin confidence–accuracy gaps:

$$\text{ECE} = \sum_{m=1}^M \frac{|b_m|}{N} |\text{acc}(b_m) - \text{conf}(b_m)|, \quad (4)$$

where $|b_m|$ is the number of samples in bin b_m , $\text{acc}(b_m)$ is the fraction of correct predictions, and $\text{conf}(b_m)$ is the mean predicted probability within that bin.

The remaining metrics follow the same bin-level discrepancy idea with specific modifications. **Maximum Calibration Error (MCE)** [19] replaces the weighted average with the worst-case bin maximum. **Classwise ECE (CECE)** [12] computes the ECE-style sum separately for each class and averages across all K classes. **Adaptive Calibration Error (ACE)** [20] replaces fixed-width bins with equal-population bins, also averaging per class. **Thresholded ACE (TACE)** [20] further restricts ACE to predictions whose confidence exceeds a threshold ϵ , reducing the influence of near-zero probabilities. Finally, the **Brier Score (BS)** [2] is a proper scoring rule computing the mean squared error between predicted probabilities and one-hot targets across all classes, providing a combined measure of calibration and sharpness.

5 Results

Here we present the results of our experiments, including calibration analysis for standard Fine-Tuning (FT), Low-Rank Adaptation (LoRA), and our proposed hyper-network-based variants.

5.1 Calibration of the Existing Methods

Table 1 presents task performance and calibration metrics for standard Fine-Tuning and LoRA fine-tuning across selected GLUE benchmarks. LoRA consistently matches or improves predictive performance relative to full Fine-Tuning, with noticeable gains on CoLA, MRPC, QNLI, and RTE, while maintaining comparable results on MNLI.

Table 1. Performance and calibration metrics across GLUE benchmarks for RoBERTa_{large}. FT denotes standard fine-tuning [16]; LoRA denotes Low-Rank Adaptation [10]. Following [24], we report Matthews Correlation Coefficient (MCC) for CoLA, F1 for MRPC, and Accuracy for other tasks. Minor score deviations from [10] are attributable to different random seeds; additionally, MRPC reports F1 (as in [16]) rather than accuracy, and RTE (FT) is initialized from MNLI-pretrained weights for fair comparison [10].

	Metric	CoLA	SST-2	QNLI	MRPC	RTE	MNLI
FT	<i>Score</i> ↑	61.68 ± _{0.73}	94.50 ± _{0.41}	92.64 ± _{0.22}	90.37 ± _{0.41}	85.32 ± _{1.37}	87.17
	ECE ↓	0.136 ± _{0.008}	0.035 ± _{0.017}	0.072 ± _{0.002}	0.111 ± _{0.023}	0.104 ± _{0.045}	0.074
	CECE ↓	0.138 ± _{0.007}	0.036 ± _{0.016}	0.072 ± _{0.002}	0.114 ± _{0.021}	0.113 ± _{0.038}	0.050
	MCE ↓	0.339 ± _{0.074}	0.277 ± _{0.242}	0.539 ± _{0.111}	0.308 ± _{0.163}	0.309 ± _{0.159}	0.202
	ACE ↓	0.134 ± _{0.009}	0.033 ± _{0.016}	0.071 ± _{0.002}	0.110 ± _{0.021}	0.103 ± _{0.039}	0.073
	TACE _{0.01} ↓	0.469 ± _{0.006}	0.461 ± _{0.013}	0.482 ± _{0.007}	0.441 ± _{0.020}	0.413 ± _{0.058}	0.492
	Brier Score ↓	0.290 ± _{0.006}	0.096 ± _{0.006}	0.145 ± _{0.004}	0.245 ± _{0.013}	0.266 ± _{0.009}	0.207
LoRA	<i>Score</i> ↑	63.94 ± _{0.21}	94.99 ± _{0.18}	93.07 ± _{0.05}	93.18 ± _{0.40}	87.61 ± _{0.21}	87.19
	ECE ↓	0.120 ± _{0.025}	0.046 ± _{0.001}	0.036 ± _{0.011}	0.088 ± _{0.012}	0.124 ± _{0.007}	0.043
	CECE ↓	0.123 ± _{0.024}	0.046 ± _{0.001}	0.037 ± _{0.011}	0.088 ± _{0.011}	0.125 ± _{0.006}	0.029
	MCE ↓	0.282 ± _{0.038}	0.340 ± _{0.111}	0.123 ± _{0.052}	0.424 ± _{0.097}	0.502 ± _{0.156}	0.257
	ACE ↓	0.114 ± _{0.024}	0.040 ± _{0.003}	0.035 ± _{0.010}	0.084 ± _{0.012}	0.114 ± _{0.002}	0.043
	TACE _{0.01} ↓	0.443 ± _{0.026}	0.477 ± _{0.007}	0.452 ± _{0.012}	0.447 ± _{0.005}	0.451 ± _{0.024}	0.432
	Brier Score ↓	0.271 ± _{0.020}	0.096 ± _{0.002}	0.114 ± _{0.004}	0.180 ± _{0.017}	0.244 ± _{0.009}	0.193

From a calibration perspective, ECE remains relatively stable across seeds for both methods, typically exhibiting low variance. CECE closely follows ECE, reflecting the predominantly binary structure of the evaluated tasks. As expected, MCE shows substantially higher variability due to its sensitivity to worst-case confidence bins. ACE aligns closely with ECE, indicating an approximately uniform distribution of samples across confidence bins.

Across tasks, LoRA demonstrates improved calibration on QNLI and MNLI, where ECE and Brier Score are consistently lower than under full Fine-Tuning. However, this behavior is not uniform. On SST-2 and RTE, Fine-Tuning achieves lower ECE values, suggesting better calibration despite slightly weaker predictive performance. The elevated values of TACE_{0.01} (TACE with threshold equal to 0.01) across both methods indicate that predictions are concentrated near extreme probabilities, leading to high-confidence outputs and increased calibration penalties when misclassifications occur.

Overall, LoRA does not uniformly improve calibration over full Fine-Tuning. Its effect is task-dependent, and improvements in predictive performance do not systematically translate into better uncertainty estimation.

5.2 Our Approach - Calibration with Hyper-Network

Table 2 reports the performance and calibration of the proposed hyper-network variants on CoLA and SST-2, compared to the LoRA baseline. We evaluate both MLP-based and Transformer-based hyper-networks, with either full generation of adapter matrices (A_{gen}) or with matrices A fixed (A_{fix}).

On CoLA, the Transformer-based hyper-network with fully generated matrices (Transformer A_{gen}) achieves average highest performance, but also shows

Table 2. We compare four hypernetwork configurations combining two architectures (MLP and Transformer) with two weight generation strategies (A_{gen} and A_{fix}), each averaged over 3 seeds. Transformer A_{gen} outperforms the LoRA baseline on CoLA, though LoRA remains stronger on SST-2, suggesting task-dependent behavior. Interestingly, Transformer A_{fix} achieves the best calibration across all runs despite not leading in Matthews Correlation Coefficient (MCC) or Accuracy.

	COLA		SST-2	
	MCC $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$
MLP A_{gen}	63.54 $\pm$ 0.31	0.116 $\pm$ 0.019	94.84 $\pm$ 0.30	0.037 $\pm$ 0.007
MLP A_{fix}	48.25 $\pm$ 13.01	0.130 $\pm$ 0.015	92.89 $\pm$ 1.41	0.047 $\pm$ 0.011
Transformer A_{gen}	64.42$\pm$1.75	0.119 $\pm$ 0.009	94.78 $\pm$ 0.08	0.040 $\pm$ 0.003
Transformer A_{fix}	60.69 $\pm$ 0.35	0.100$\pm$0.010	94.56 $\pm$ 0.08	0.028$\pm$0.004
LoRA	63.94 $\pm$ 0.21	0.120 $\pm$ 0.025	94.99$\pm$0.18	0.046 $\pm$ 0.001

high variability across different seeds. It surpasses LoRA fine-tuning. However, its calibration remains similar to LoRA, indicating no inherent calibration advantage from full matrix generation. When A matrices are fixed (Transformer A_{fix}), MCC value decreases, but ECE improves, representing the best calibration among all evaluated configurations. We hypothesize that this calibration improvement is caused by a reduction in effective degrees of freedom - fixing A matrices constrains the adaptation space, acting as an implicit regularizer that prevents the model from becoming overconfident.

The MLP-based variants follow the same qualitative pattern but exhibit reduced stability. While MLP A_{gen} maintains competitive performance, fixing A matrices leads to a substantial drop in performance without improving calibration relative to the Transformer-based fixed configuration.

On SST-2, LoRA achieves the highest predictive accuracy. Both hyper-network variants with generated matrices remain competitive. Again, freezing matrices A results in a small but consistent decrease in accuracy. However, the Transformer-based fixed configuration yields the lowest ECE, significantly outperforming both LoRA and the fully generated variant.

Contrary to our initial hypothesis, fully generated hyper-network adapters do not systematically improve calibration over LoRA, despite occasionally improving task performance (notably on CoLA). Instead, calibration improvements emerge primarily when matrices A are fixed. This constraint introduces a structured transformation of the adapter input space, limiting overconfident predictions and acting as a form of implicit regularization. The improvement in calibration is accompanied by a clear trade-off in predictive performance, particularly pronounced on CoLA.

Additionally, as shown on Fig. 2, we observe that extended training consistently leads to worsening calibration across all configurations, even when task metrics continue to improve. This behavior suggests progressive overfitting to the training objective, resulting in sharper predictive distributions and degraded uncertainty estimation.

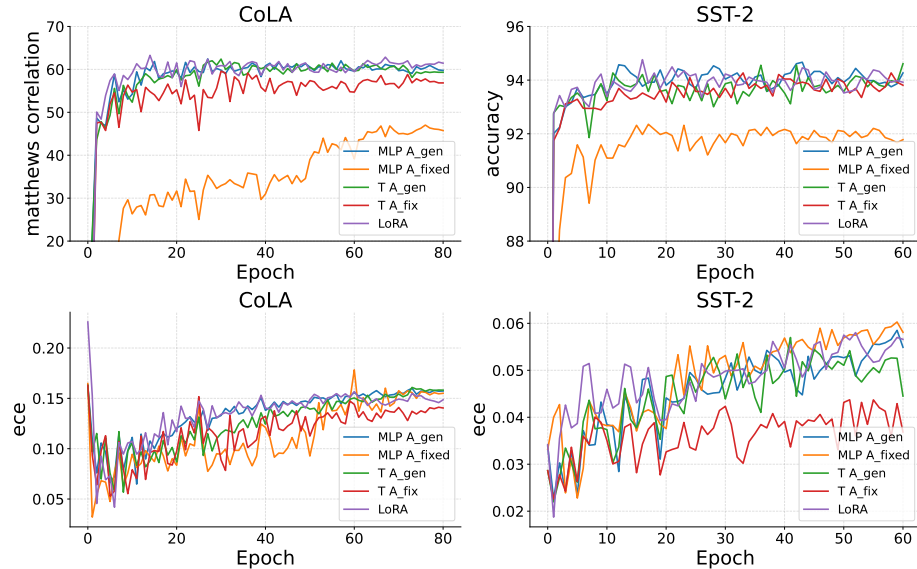


Fig. 2. Evaluation results on CoLA and SST-2 benchmarks, reported as Matthews Correlation Coefficient and accuracy (top row) alongside Expected Calibration Error (bottom row), averaged across 3 independent random seeds. A_{gen} means both matrices A and B are generated, and A_{fix} means matrices A are fixed. LoRA [10] is included as a baseline. Fixing matrix A improves model calibration, albeit at the cost of task performance across both datasets.

6 Conclusion

In this work, we investigated parameter-efficient adaptation mechanisms as a novel approach to calibration for transformer-based language models. We analyzed standard Fine-Tuning and LoRA fine-tuning [10] applied to RoBERTa [16], and introduced a hyper-network-based variant in which low-rank update matrices are generated conditionally on transformer layer identity.

Our evaluation across multiple GLUE benchmarks [24] yields three main findings. First, LoRA provides calibration comparable to full Fine-Tuning while retaining parameter efficiency, though calibration improvements remain task-dependent. Second, hyper-network-generated low-rank factors yield calibration broadly similar to standard LoRA, suggesting structural cross-layer coupling alone is insufficient for systematic confidence correction, but also our proposed Transformer-based hyper-network LoRA variant (Transformer A_{gen}) showed promising results, by outperforming standard LoRA on the CoLA benchmark. Third, freezing all of the A matrices (A_{fix}) modestly improves calibration at the cost of task performance, revealing a tension between representation stability and predictive sharpness. We additionally provide a unified, reproducible implementation of six calibration metrics (ECE, CECE, MCE, ACE, TACE, Brier Score),

contributing toward more systematic evaluation standards in transformer calibration research.

Further Work Future work should investigate the mechanism behind the A_{fix} calibration improvement — whether it stems from reduced flexibility, additional noise, or modified optimization dynamics. The CoLA advantage of the Transformer hyper-network motivates further study of this architecture. Extending evaluation to multi-class and out-of-distribution benchmarks would clarify whether observed improvements generalize beyond binary GLUE tasks.

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